Test Documentation. Challenges API Tests

Overview

This document provides comprehensive documentation for the test suite covering the Challenges API endpoints. The suite contains **37 tests** organized into 10 logical groups, achieving near-complete coverage of the challenge-related functionality.

Test Categories

1. Challenge Creation Tests (`TestChallengeCreation`)

Test Name	Description
`test_create_challenge_success`	Verifies successful creation of a challenge with multiple exercises; ensures creator is auto-joined and receives a join code
`test_create_challenge_with_weekly_schedule`	Tests challenge creation with weekly scheduling (Monday, Wednesday, Friday); validates schedule_days array storage
`test_create_challenge_with_open_end`	Ensures challenges can be created without an end_date (open-ended challenges)
`test_create_challenge_fails_without_exercises`	Validates that challenges require at least one exercise (422 validation error)
`test_create_challenge_fails_with_duplicate_exercises`	Ensures the same exercise cannot be

Test Name	Description
	added twice to one challenge
`test_create_challenge_fails_with_invalid_exercise`	Tests handling of non-existent exercise_id (400 Bad Request)
`test_create_challenge_fails_with_invalid_schedule_days`	Validates schedule_days are in range 1-7 for weekly schedules
`test_create_challenge_fails_with_end_date_before_start`	Ensures end_date cannot precede start_date
`test_create_challenge_without_auth`	Verifies authentication is required to create challenges (401 Unauthorized)

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2. Challenge Detail Tests (`TestChallengeDetail`)

Test Name	Description
`test_get_challenge_detail`	Validates retrieval of complete challenge details including exercises, participants count, and joined status
`test_get_challenge_detail_not_found`	Ensures 404 response for non-existent challenge IDs
`test_get_challenge_detail_join_code_only_for_creator`	Confirms join_code is visible only to the challenge creator, not to other users

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3. Challenge Edit Tests (`TestChallengeEdit`)

Test Name	Description
`test_edit_challenge_name`	Verifies successful editing of name and description fields
`test_edit_challenge_by_non_creator`	Ensures only the creator can edit a challenge (403 Forbidden)
`test_edit_challenge_not_found`	Tests 404 response when attempting to edit non-existent challenge

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4. Join/Leave Tests (`TestJoinLeave`)

Test Name	Description
`test_join_by_code_success`	Tests successful challenge joining using a valid join_code
`test_join_by_code_invalid`	Ensures 404 response for invalid join codes
`test_join_already_joined`	Validates users cannot join the same challenge twice (409 Conflict)
`test_join_inactive_challenge`	Ensures users cannot join archived/completed challenges (409 Conflict)
`test_leave_challenge`	Tests successful exit from a challenge; verifies participation is removed
`test_leave_not_participant`	Ensures 404 response when non-participant attempts to leave

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5. Session Submission Tests (`TestSessionSubmission`) Core Logic

Test Name	Description
`test_submit_session_closes_exercise`	Verifies an exercise is marked closed when clean_reps reach or exceed goal

Test Name	Description
`test_submit_session_closes_day`	Tests full day completion when all exercises in a challenge are closed; validates streak increment
`test_submit_session_fails_if_not_active`	Ensures sessions cannot be submitted to archived challenges (409 Conflict)
`test_submit_session_fails_if_not_participant`	Validates only participants can submit sessions (403 Forbidden)
`test_submit_session_fails_with_invalid_exercise`	Ensures exercises must belong to the challenge (404 Not Found)
`test_submit_session_accumulates_today`	Tests multiple sessions per day accumulate; first session may not close, second does

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6. Leaderboard Tests (`TestLeaderboard`)

Test Name	Description
`test_leaderboard_returns_sorted`	Validates leaderboard sorting by days_completed, challenge_streak, total_clean_reps, and joined_at
`test_leaderboard_requires_auth`	Ensures authentication is required for leaderboard access (401 Unauthorized)

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7. Presets Tests (`TestPresets`)

Test Name	Description
`test_get_presets`	Verifies retrieval of preset/system challenges list; returns empty list or presets

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8. Me Routes Tests (`TestMeRoutes`)

Test Name	Description
`test_me_profile`	Validates user profile endpoint returns correct user data including streak information
`test_my_challenges_active`	Filters and returns only active challenges for the current user
`test_my_challenges_archived`	Filters and returns only archived challenges for the current user

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9. Archive Tests (`TestArchive`)

Test Name	Description
`test_archive_challenge_by_creator`	Allows challenge creator to archive their challenge
`test_archive_challenge_by_non_creator`	Prevents non-creators from archiving (403 Forbidden)
`test_archive_already_archived`	Ensures idempotent archiving (subsequent archive calls return success)

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10. Exercises List Tests (`TestExercisesList`)

Test Name	Description
`test_list_exercises`	Verifies the exercise catalog returns at least the 3 seeded exercises

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Test Environment

Component	Configuration
Database	SQLite in-memory (per test isolation)
Authentication	JWT tokens generated per test
Seed Data	3 exercises: Squats, Pushups, Plank
Test Runner	pytest 9.1.1
HTTP Client	FastAPI TestClient (httpx)

Coverage Summary

Category	Tests	Status
Challenge Creation	9	All Passing
Challenge Detail	3	All Passing
Challenge Edit	3	All Passing
Join/Leave	6	All Passing
Session Submission	6	All Passing
Leaderboard	2	All Passing
Presets	1	All Passing
Me Routes	3	All Passing
Archive	3	All Passing
Exercises List	1	All Passing
Total	**37**	**37/37 Passing**

Running the Tests

```shell
# Run all tests with verbose output
pytest tests/test_challenges.py -v

# Run specific test class
pytest tests/test_challenges.py::TestSessionSubmission -v

# Run with coverage report
pytest tests/test_challenges.py --cov=app

# Run failing tests only
pytest tests/test_challenges.py --lf
```

Notes for Developers

1. **Database Isolation:** Each test runs in a fresh database with seeded exercises
2. **Date Handling:** All creation tests use tomorrow's date to bypass validation
3. **Authentication:** Each test generates fresh JWT tokens
4. **Session Accumulation:** Multiple sessions in a day sum clean_reps until goal is met
5. **Streak Logic:** User streak increments only once per calendar day